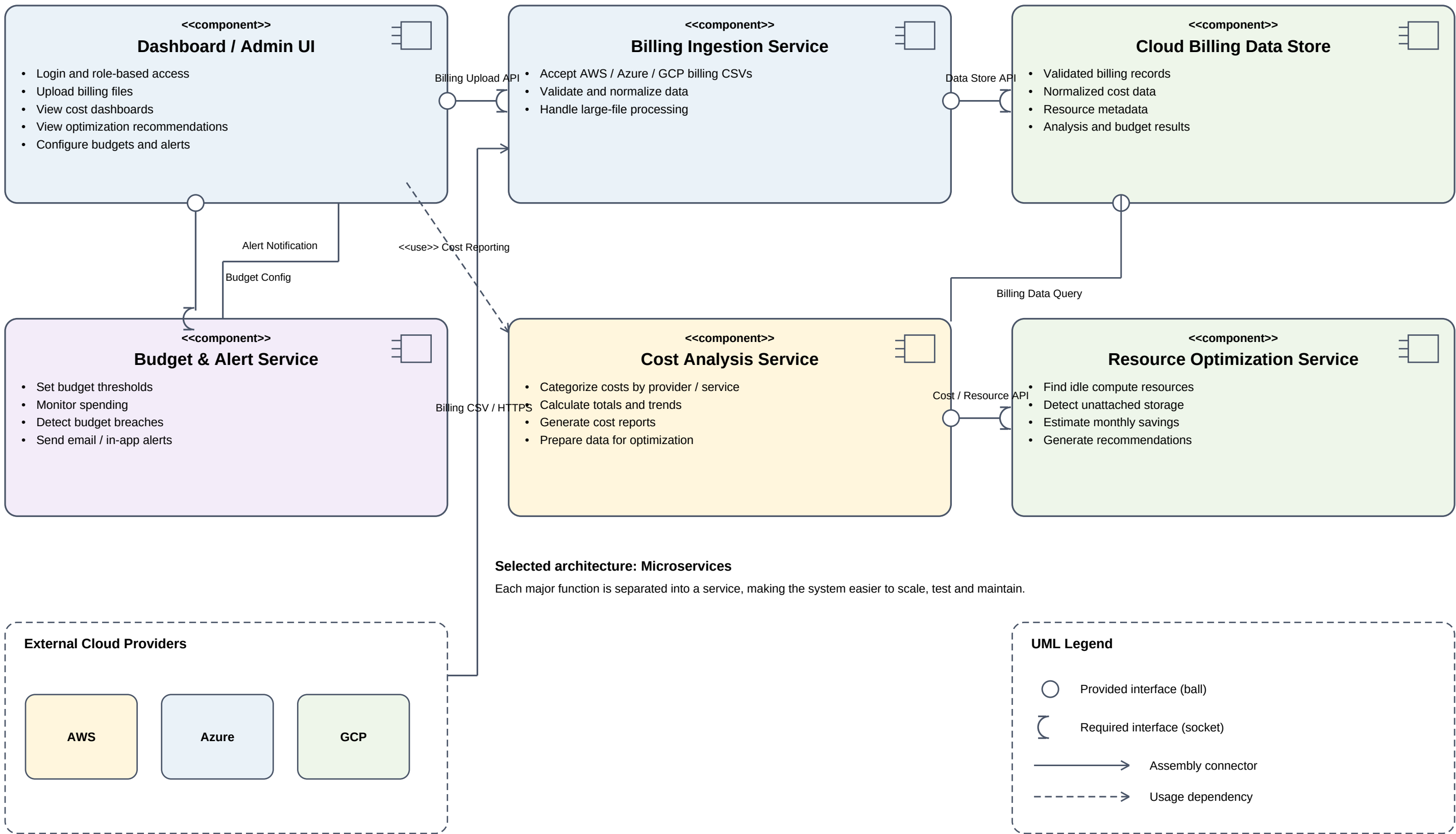




Software Engineering Lab 3 - Architecture Justification

Problem No.: 45 | Name: Sreekar Bandi | SRN: PES1UG24AM282

System: Multi-Cloud Asset Cost Optimization Portal

Architecture Selection

We selected **Microservices Architecture** for the Multi-Cloud Asset Cost Optimization Portal because the system can be divided into separate services for ingestion, analysis, optimization, budgeting, and the user interface.

Justification

1. The microservices approach allows billing ingestion and cost analysis to scale independently, which is useful because these services may receive heavier workloads than the dashboard and must handle large AWS, Azure, and GCP billing inputs efficiently.
2. It also provides fault isolation, so a problem in one service such as budget alerts or optimization analysis can be contained without making the entire cost-optimization portal unavailable.
3. From a security perspective, each service can enforce appropriate authentication and role-based access controls at its boundary, while data can be protected using encryption during transmission and storage as required by the project.
4. From a performance perspective, different services can process work in parallel and be scaled according to demand, helping the dashboard remain responsive and supporting the target of updating relevant results within about two minutes.
5. Overall, microservices is a practical fit for this project because its independent services make the portal easier to scale, maintain, secure, and extend as more cloud billing sources and optimization features are added.

Conclusion

Therefore, Microservices Architecture is the most suitable choice for Problem 45, as it directly supports the portal's functional needs while giving the system a clear structure for scalability, reliability, security, and performance.